

SOCAR WIN OPEN INNOVATION PROGRAM • INVESTOR PITCH

ASTEM

AI-Powered Industrial & Urban Waste Heat Recovery

Transforming Industrial Effluent & Infrastructure Waste Heat into Low-Carbon Energy for SOCAR



Refineries & Petrochemical



AI Heat-Pump Upgrade



Metro & Transit



Low-Carbon Cities

THE TEAM

ASTEM Core Team

Six-member execution team aligned for SOCAR pilot delivery



TEAM LEADER

Isa Ibadov

Strategy & SOCAR partnerships.



TECHNICAL MEMBER

Shukurov Shukur

System design and installation specifications.



FUNDER

Rufat Jabrayilli

Business model and HPA structuring.



RESEARCH TEAM

Ziya Aliyev • Shargiyya Qafarzade • Aysel Sardarova

Aspen HYSYS modeling and telemetry.

The Industrial & Urban Heat Paradox



Assets Generate Massive Low-Grade Heat

SOCAR refineries and petrochemical plants discharge heat continuously through cooling towers and process effluent. Metro tunnels add dense, high-frequency passenger heat.



Valuable Energy Lost to the Atmosphere

That 18–40°C+ thermal energy is vented or dumped as waste — while nearby boilers keep burning natural gas for feedwater heating.

Key Point

Heydar Aliyev Refinery and SOCAR Polymer discharge massive low-grade waste heat (18°C–40°C+) through cooling towers and liquid process effluent into the atmosphere, while site boilers burn natural gas to heat feedwater — the same paradox seen underground, where Baku Metro tunnels stay 18°C–30°C year-round and vent that energy through ventilation shafts.

The Concept: AI-governed heat pump capture converts this low-grade waste into 60°C–90°C hot water for industrial boiler pre-heating.

Capturing SOCAR's Wasted Industrial & Urban Energy



THE PARADOX

Refinery cooling towers, process effluent, and metro tunnels all sit at 18°C–40°C+ year-round — energy currently wasted while nearby boilers burn fossil fuels for heat.



THE CONCEPT

AI-governed Water-to-Water and Air-to-Water heat pumps capture this stable low-grade heat at industrial and transit nodes before it escapes.



THE OPPORTUNITY

Transform 'waste heat' into a sustainable 60°C–90°C supply for boiler feedwater pre-heating, district heat, and facility hot water.

Global precedent: modeled on London Bunhill 2 (500+ homes heated) and Warsaw M2 heat-recovery systems.

Two Thermal Sources, One AI-Governed Network

Steady industrial process heat and dense urban transit heat combine into a resilient, always-on thermal supply.



SOCAR Industrial Assets

- Continuous, high-volume discharge — refineries and petrochemical plants run 24/7
- 18°C–40°C+ cooling-tower and effluent streams
- Heydar Aliyev Refinery • SCIP • SOCAR Polymer • SOCAR Carbamide



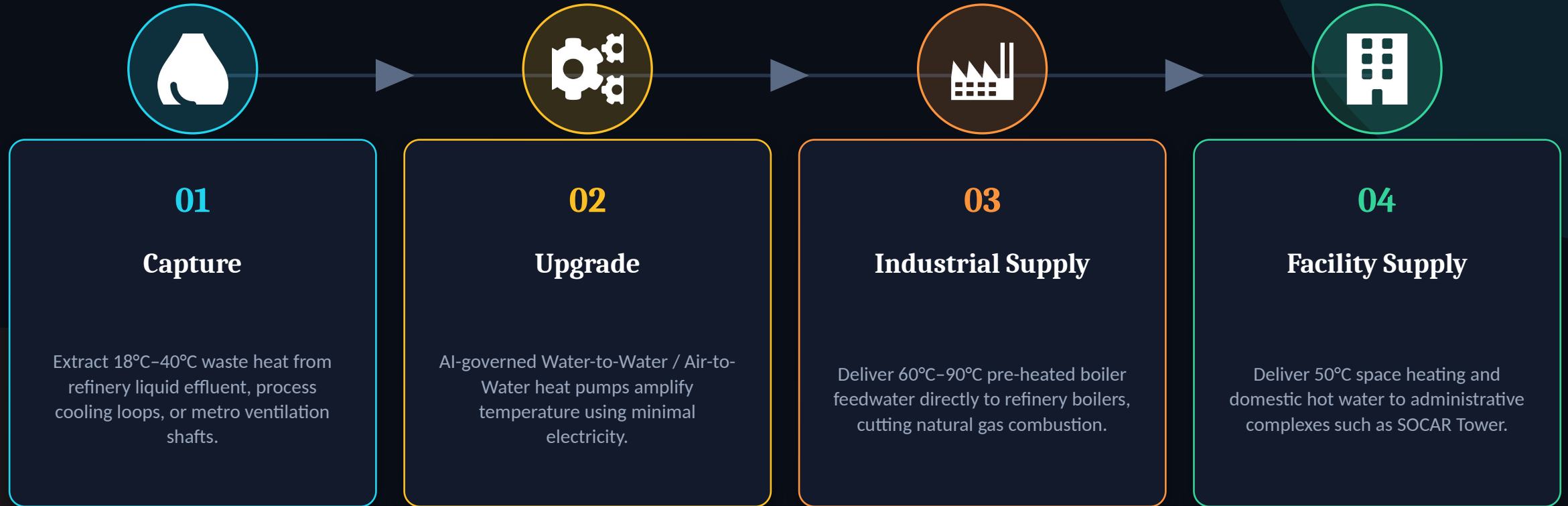
Baku Metro Transit Network

- Peak commute-hour passenger heat across 27 measured stations
- 8.3 million passengers/station — more than double London's density
- 18°C–30°C stable tunnel air year-round

500,000+ kW

Combined recoverable thermal potential across industrial and urban assets

How It Works



Asset Heat Mapping & Impact Metrics



500,000+ kW
Recoverable Thermal Potential



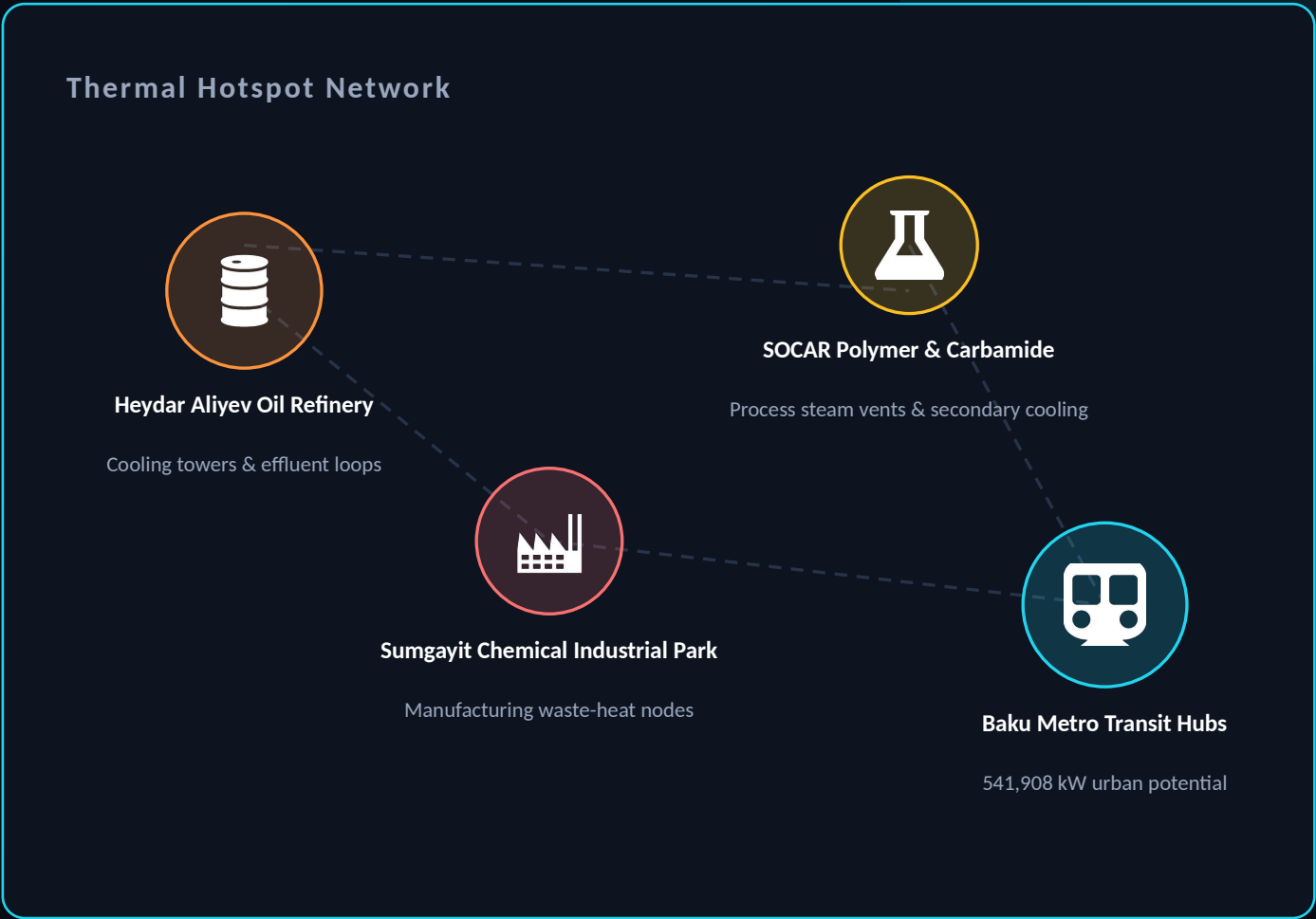
2,000,000+ MWh
Annual Clean Energy Recycled



400,000+ Tons
Scope 1 CO₂ Emission Reduction



Millions of m³
Natural Gas Offset Annually



Technology Efficiency & AI Engine

1 kWh Electricity + **2.8 kWh Waste Heat** = **3.8 kWh Usable Thermal Output** (COP 3.8+)

Standard ASHP

Pulling from 0°C freezing winter outside air results in high electricity demand, low efficiency, and high cost.

Astem Industrial System

Pulling from stable 20°C–40°C industrial effluent and tunnel air results in low electricity demand, high efficiency (COP 3.8+), and high profitability.



AI Advantage

Machine-learning algorithms ingest real-time flow rate, temperature, and industrial steam demand to optimize compressor speeds — without disrupting primary refinery operations.

Operational Utility: Seasonal Modes



Winter Mode

- Refinery boiler feedwater pre-heating — direct natural gas OPEX reduction
- Facility space heating for administrative complexes
- Recovers heat efficiently from stable underground and process sources

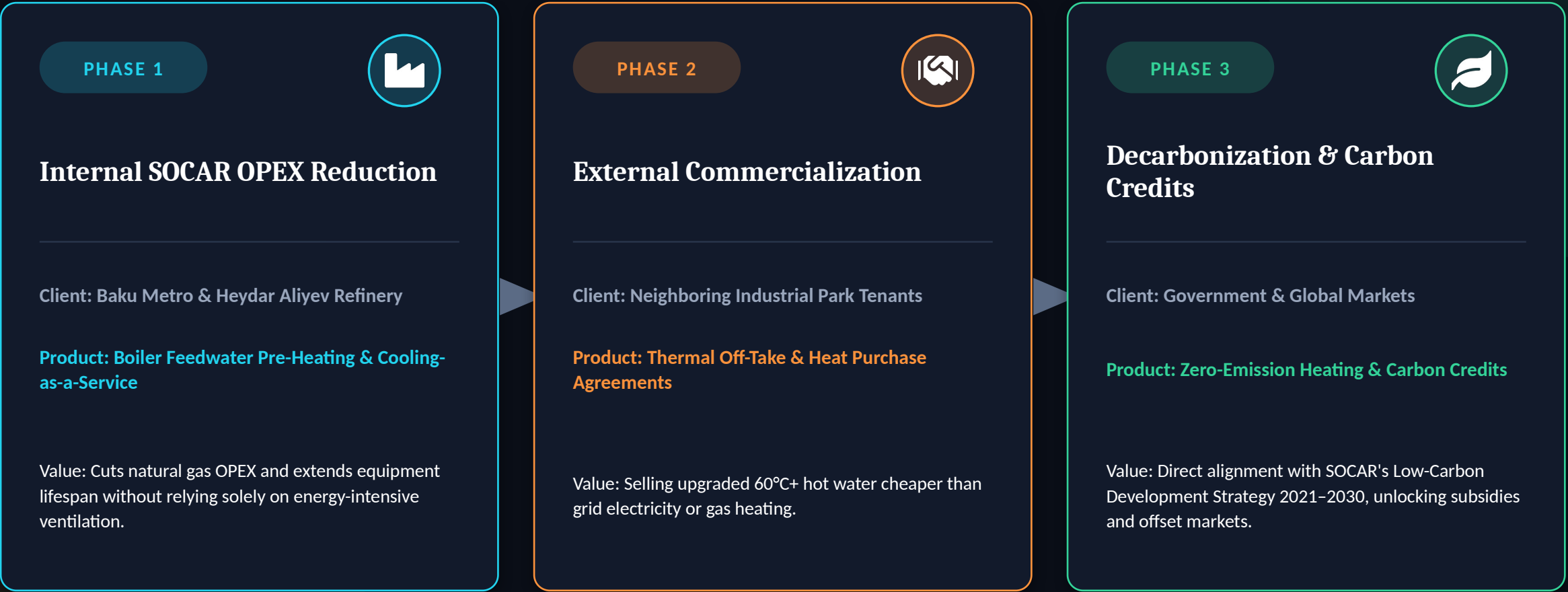


Summer Mode

- Carbon-free HVAC cooling for SOCAR offices
- Asset thermal management — lowers ambient temperatures around electrical and refinery equipment
- Extends machinery lifespan while cooling stations and heating water

Prioritizes operational continuity and equipment longevity year-round.

A Strategic Roadmap to Profitability



Financial Feasibility — 1.8 MW Industrial Pilot



1.8 MW

Thermal Capacity



11.9 GWh

Annual Energy Saved



476,000 AZN

Annual Cost Savings (~\$280,000 USD)



4.9–7 Years

Payback Period

Fund the SOCAR Industrial Pilot.
Validate the AI digital twin.
Scale across SOCAR's petrochemical portfolio.
Transform urban and industrial energy forever.

Deploy a 150 kW – 1.8 MW unit at a refinery stream or SOCAR Tower to validate performance before portfolio-wide scale-up.



Join the Astem Project